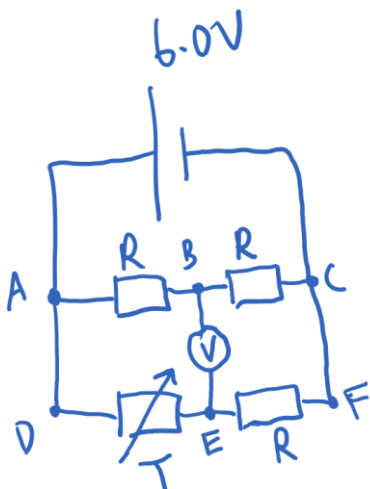


$$W = VQ = VI t = 16 \times 1.5 \times 0.2 = 5.4 \text{ J}$$

22

B

Redraw the circuit diagram.



Initially, resistance of $T = R$.

Pd across each resistor = pd across $T = 3\text{V}$.

Potential at $B = \text{Potential at } E = 3\text{V}$, so pd across $BE = 0$, so voltmeter reading = 0.

When resistance of $T = 2R$,

	Potential at B is still 3V. Potential at E = $6 \times \frac{R}{2R + R} = 2 \text{ V}$. Potential across BE = $3 - 2 = 1 \text{ V}$, so voltmeter reading = 1V. Change in voltmeter reading = $1 - 0 = 1\text{V}$
23	C